DP Graded Quiz M4

Question 1

When implementing the GBM equation from the notes on the simulation of a stock price, you use the following command:

np.random.randn(500, 252)

How many points (prices) are generated in each path?

- 500x252
- 252
- 500
- 251

Question 2

In the following Vasicek (1977) type model...

$$dr_t = \lambda(\phi - r_t)dt + \sigma dZ_t$$

- $\frac{\sigma^2}{2\lambda}$ is the instantaneous volatility of the model.
- $\frac{\sigma^2}{2\lambda}$ is the long-term mean of interest rates.
- $\frac{\sigma^2}{2\lambda}$ is the long-term variance.
- dZ_t follows a N(0,1) distribution.

Question 3

(**Note:** Use Python to solve this question.) Price a European Put option using a Monte-Carlo method under Black-Scholes GBM dynamics. The strike price of this option is $122, with the current stock price at $118.75. The risk-free rate is at 1.5% and the option will expire in 3 months. Volatility is 25%. Make sure you use a random seed equal to 2. What is the price of the option after 3,000 simulations? (Please, use np.random.normal( ) for random number generation. Also, select the answer closer to your result.)

- $4.595
- $3.25
- $13.165
- $7.586

Question 4

You just purchased a European Call option. The strike price of the option is $35, and there are 2 months left until maturity. The current price of the underlying stock price is $32.5. The volatility for this stock is 45% annually, while the risk-free rate is at 1%. Calculate the sensitivity of the option price to the risk-free rate under the Black-Scholes framework. Feel free to use the provided Python code for this.

- -6.587
- -3.998

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- 1.825
- -6.936

Question 5

Suppose $W_{1,t}$ and $W_{2,t}$ are independent Brownian motions with variance (σ^2) parameters 0.0625 and 0.09, respectively. What will be the volatility (σ) of process $Z_t = 0.75W_{1,t} + 0.25W_{2,t}$?

- 20.19%
- 12.18%
- 6.94%
- 26.34%

Question 6

(**Note:** Use Python to solve this question.) Consider the following Vasicek (1977) model:

$dr = \kappa(\theta - r)dt + \sigma dz$, with $dz = \sqrt{dt}z$, and $Z \sim N(0,1)$

Use Monte Carlo to simulate 25 paths of the interest rate in 3 months using 250 steps and a random seed equal to 42. Also use the following parameters:

$\kappa = 0.15$

$\theta = 0.0255$

$\sigma = 0.0085$

$r_0 = 0.0285$

What is, among the 25 paths simulated, the lowest interest rate simulated in 3 months?
(Please, use np.random.normal( ) for random number generation. Also, select the answer closer to your result.)

- 2.8550%
- 2.8284%
- 1.0139%
- 1.884%

Question 7

(**Note:** Use Python to solve this question.) Price a European Call option using a Monte-Carlo method under Black-Scholes GBM dynamics. The strike price of this option is $122, with the current stock price at $118.75. The risk-free rate is at 1.5% and the option will expire in 3 months. Volatility is 25%. Make sure you use a random seed equal to 2. What is the price of the option after 3,000 simulations? (Please, use np.random.normal( ) for random number generation. Also, select the answer closer to your result.)

- $10.987
- $7.810
- $4.595
- $3.25

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Question 8

Suppose $\pi = \alpha S^2$ represents a derivative claim. Which of the following is the expression for the Black-Scholes-Merton PDE?

- $\alpha S^2(\sigma^2 + r) = 0$
- $2\alpha S^2(\sigma^2 + r/2) = 0$
- $2\alpha S^2(\sigma^2 + r) = 0$
- $\alpha S^2(\sigma^2 + r/2) = 0$

Question 9

In the following generic GBM equation:

$dS = S(\mu dt + \sigma dW_t)$

- σdW_t is commonly known as "the drift" term
- μ accounts for underlying process volatility
- dW_t follows a Normal distribution
- dW_t follows a Poisson distribution

Question 10

Suppose r_T is the continuously compounded interest rate. Then, the expression for the discount factor at time 0, $P_0(T)$, is:

- $P_0(T) = e^{-r_T T}$
- $P_0(T) = \frac{1}{(1 + r_T)^T}$
- $P_0(T) = e^{-r_T/T}$
- $P_0(T) = (1 + r_T)^{-T}$

Question 11

Consider the following GBM process given by the following SDE:

$dX_t = X_t(\mu dt + \sigma dW_t)\$\$. Suppose\$Y_t = logX_t\$. ApplyingIt\hat{o}sruleandsolvingtheequatic$

- $Y_t = Y_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$
- $Y_t = X_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$
- $X_t = X_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t]$
- $X_t = X_0 + (\mu - \frac{1}{2}\sigma)t + \sigma W_t]$

Question 12

Suppose $\pi = S^{\frac{\alpha}{2}}$ represents a derivative claim. Which of the following is the expression for the Black-Scholes-Merton PDE?

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- $(\frac{\alpha}{2})(\frac{1}{2}\sigma^2\frac{\alpha}{2}+r)S^{\alpha/2}=0$
- $(\frac{\alpha}{2}-1)(\frac{1}{2}\sigma^2\frac{\alpha}{2}+r)S^{\alpha/2}=0$
- $(\alpha-1)S^{\alpha}(r+\frac{1}{2}\sigma^2\alpha)=0$
- $(\alpha-1)S^{\alpha/2}(r+\frac{1}{2}\sigma^2\alpha)=0$

Question 13

The current price of a stock is $S_0 = 230$. Annual return and volatility for this stock has historically been 7% and 26%, respectively. What will be the expected change in the price tomorrow if we model it with 1 simulation of the standard GBM?

- Assume there are 255 days in a year.
- Use Python to solve this. You should set the random seed equal to 1 when generating random numbers.

- $4.186
- $16.100
- $1.624
- $6.146

Question 14

In the following Vasicek (1977) type model...

$$dr_t = x(y - r_t)dt + \sigma dW_t$$

- Parameter x proxies for the long-term mean of interest rates.
- Parameter y proxies for the speed of mean-reversion.
- Parameter $\frac{\sigma^2}{2x}$ proxies for the long-term variance.
- Parameter $\frac{\sigma^2}{2y}$ proxies for the long-term variance.

Question 15

Regarding the power of Monte-Carlo methods...

- They do not require an analytical expression (model) to run simulations on.
- They are of no use when there is no closed-form solution to the model equation.
- They yield more accurate values than analytical methods.
- They are more accurate as the number of simulations increases.

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1 2
5 6
9 10
13 14
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Question 16

In the Black-Scholes model, we assume that asset returns are distributed...

- using Poisson distribution
- No particular distribution is assumed
- Lognormally
- Normally

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